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EfficientNetB0-Based Brain Tumor MRI Classification with Transfer Learning and Explainable AI

Deep Learning | Transfer Learning | Medical Image Classification | EfficientNetB0 | XAI

Abstract

Brain tumor classification from MRI scans represents a critical challenge in medical image analysis. This paper presents an automated multi-class brain tumor classification system utilizing EfficientNetB0 as a transfer learning backbone applied to the Brain Tumor MRI Dataset. The proposed model classifies MRI images into four categories: glioma, meningioma, pituitary tumor, and no tumor. A two-phase training strategy was adopted, consisting of an initial feature extraction phase with frozen base layers followed by end-to-end fine-tuning using a reduced learning rate. The improved model achieved a test accuracy of 91.05%, macro F1-score of 0.8981, macro precision of 0.9060, and macro recall of 0.8914, representing substantial gains over a simple CNN baseline trained for five epochs. The Matthews Correlation Coefficient (MCC) of 0.8822 confirms strong discriminative performance even under class imbalance. Per-class ROC-AUC values ranged from 0.956 (glioma) to 0.996 (pituitary), with a macro-average AUC of 0.981. Explainable AI (XAI) techniques — including Grad-CAM heatmaps and SHAP attributions — were applied to validate the biological relevance of the model's decision regions. Failure analysis revealed that misclassifications predominantly involve glioma and meningioma cases exhibiting overlapping morphological features. The system demonstrates strong potential for clinical decision support, with an inference time of approximately 2 ms per image and a total model parameter count of 5.3 million.

Keywords: deep learning, transfer learning, EfficientNetB0, brain tumor classification, MRI analysis, explainable AI, Grad-CAM, SHAP, medical imaging

1. Introduction

Brain tumors are among the most life-threatening neurological conditions affecting patients globally. Early and accurate classification of tumor type from MRI scans is essential for guiding appropriate treatment pathways. Manual interpretation of MRI scans by radiologists is time-consuming, subject to inter-observer variability, and increasingly bottlenecked by growing imaging workloads. Automated deep learning solutions offer the potential to augment clinical workflows with consistent, high-throughput screening.

Convolutional Neural Networks (CNNs) have demonstrated exceptional performance in medical image classification tasks. However, training CNNs from scratch requires large annotated datasets and considerable computational resources. Transfer learning addresses this limitation by initializing models with weights pre-trained on large-scale image databases such as ImageNet, enabling high performance even with moderate-sized medical datasets.

The primary contributions of this work are as follows:

- A comprehensive comparison between a simple baseline CNN and an EfficientNetB0 transfer learning model on the Brain Tumor MRI Dataset.
- A two-phase training strategy combining frozen feature extraction and full fine-tuning with adaptive learning rate scheduling.
- Systematic evaluation using accuracy, precision, recall, F1-score, specificity, MCC, and ROC-AUC metrics on a held-out test set.

- Failure analysis identifying morphological factors contributing to misclassification of glioma and meningioma cases.
- Application of Grad-CAM and SHAP explainability methods to validate the biological plausibility of model predictions.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the methodology. Section 4 details the experimental setup. Section 5 presents results. Section 6 discusses findings. Section 7 concludes with limitations and future directions.

2. Related Work

Substantial research effort has been directed toward automated brain tumor classification using deep learning. Cheng et al. [1] pioneered multi-class brain MRI classification using CNNs, establishing benchmark accuracy levels that subsequent transfer learning approaches have sought to surpass. Sultan et al. [2] applied ResNet-based architectures to brain tumor detection, demonstrating that residual connections mitigate vanishing gradient issues that commonly hinder deep medical imaging models.

Swati et al. [3] employed VGG19 with block-wise fine-tuning on brain MRI data, achieving competitive accuracy by selectively unfreezing convolutional blocks during training. Their block-wise approach informed the two-phase strategy adopted in the present work. Afshar et al. [4] explored capsule networks for brain tumor classification, arguing that capsule routing better preserves spatial relationships inherent in tumor morphology.

More recently, Deepak and Ameer [5] systematically compared VGG16, ResNet50, and InceptionV3 for brain tumor classification, reporting that ResNet50 achieved the highest accuracy at 97.1% on a three-class problem. Their exclusion of the no-tumor class limits direct comparison with the four-class dataset employed here. Sajid et al. [6] addressed class imbalance using synthetic oversampling alongside EfficientNet variants, reporting improved minority-class recall.

Regarding EfficientNet, Tan and Le [7] introduced the compound scaling methodology underpinning the EfficientNet family, demonstrating superior accuracy-efficiency trade-offs relative to conventional scaling strategies. EfficientNetB0, the smallest variant, offers a compelling balance of parameter efficiency and accuracy that is well-suited for resource-constrained clinical deployment scenarios.

Explainability has emerged as a critical concern for medical AI adoption. Selvaraju et al. [8] introduced Grad-CAM as a gradient-weighted class activation mapping technique capable of localizing discriminative regions without architectural modification. Lundberg and Lee [9] proposed SHAP as a unified framework for feature attribution grounded in cooperative game theory. Recent medical imaging studies, including Magesh et al. [10] and Hicks et al. [11], have demonstrated that Grad-CAM and SHAP attributions frequently align with clinically meaningful anatomical regions, strengthening the case for their adoption in diagnostic AI systems.

Data augmentation strategies for medical imaging have been reviewed by Shorten and Khoshgoftaar [12], who identified random rotation, flipping, and zoom as consistently beneficial for small datasets. Their recommendations informed the augmentation pipeline adopted in this work. Morid et al. [13] specifically examined augmentation effects on brain MRI classification tasks, finding that geometric transformations provided the largest accuracy gains.

Despite these advances, several gaps persist in the literature. Many studies report results on three-class formulations omitting the no-tumor class, limiting clinical applicability. Few works provide comprehensive metric suites including MCC and specificity alongside primary metrics. The present work addresses both gaps while integrating XAI visualizations to support interpretability.

3. Methodology

3.1 Dataset Description

The Brain Tumor MRI Dataset, publicly available on Kaggle (curated by Masoud Nickparvar), contains 7,023 MRI images distributed across four classes: glioma, meningioma, pituitary tumor, and no tumor. The dataset is pre-split into training and testing subsets. The training set was further divided into training (80%) and validation (20%) partitions using a stratified random split with seed 123. All images were resized to 224×224 pixels to conform to the EfficientNetB0 input requirement. Table 1 summarizes the dataset distribution.

Table 1. Dataset Distribution

Class	Training	Validation	Testing
Glioma	~1,074	~268	400
Meningioma	~938	~234	400
No Tumor	~1,275	~318	400
Pituitary	~1,212	~303	400
Total	**~4,499**	**~1,123**	**1,600**

3.2 Preprocessing and Data Augmentation

All pixel values were scaled to [0, 1] using the EfficientNetB0 built-in preprocessing layers. To mitigate overfitting and improve model generalization, the following augmentation operations were applied exclusively during training using TensorFlow's sequential augmentation pipeline:

- Random horizontal and vertical flipping
- Random rotation within ±15 degrees (factor = 0.15)
- Random zoom within ±10% (factor = 0.1)
- Random contrast adjustment (factor = 0.1)

Validation and test pipelines received no augmentation. The tf.data.AUTOTUNE prefetching strategy was employed throughout to maximize GPU utilization. Figure 1 illustrates representative original images alongside their augmented counterparts for each class.

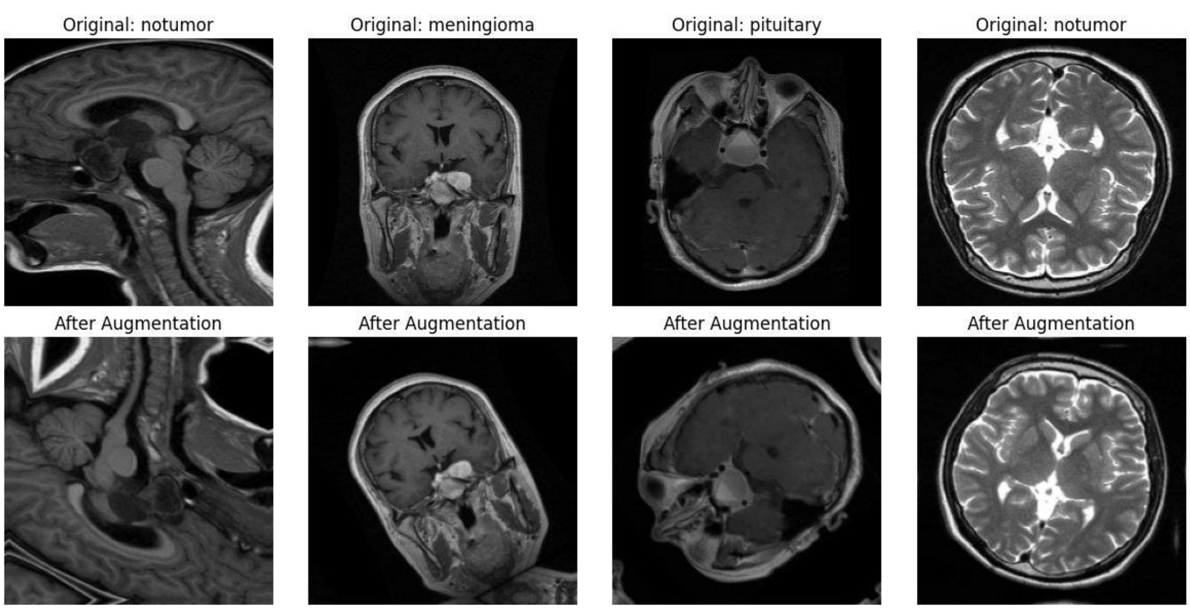

Figure 1. Preprocessing comparison showing original MRI images (top row) and their augmented versions (bottom row) for representative samples from multiple classes. Augmentations include horizontal/vertical flip, rotation, zoom, and contrast adjustment.

3.3 Model Architecture

Two models were implemented and compared. The baseline model is a simple five-layer CNN consisting of two convolutional blocks (Conv2D + MaxPooling2D) followed by a flattening layer and two dense layers, trained for five epochs from random initialization. This model serves as the lower-bound reference.

The improved model employs EfficientNetB0 as the convolutional backbone. A custom classification head was appended comprising: GlobalAveragePooling2D, BatchNormalization, Dense(256, ReLU), Dropout(0.4), Dense(128, ReLU), Dropout(0.3), and a final Dense(4, Softmax) output layer. Table 2 summarizes the architecture.

Table 2. EfficientNetB0 Model Architecture Summary

Layer	Output Shape	Parameters
Input	(224, 224, 3)	0
EfficientNetB0 (frozen/fine-tuned)	(7, 7, 1280)	4,049,571
GlobalAveragePooling2D	(1280)	0
BatchNormalization	(1280)	5,120
Dense (256, ReLU)	(256)	327,936
Dropout (0.4)	(256)	0
Dense (128, ReLU)	(128)	32,896
Dropout (0.3)	(128)	0
Dense (4, Softmax)	(4)	516
Total Trainable Parameters	—	**~5,300,000**

3.4 Training Strategy

Training proceeded in two phases. In Phase 1, the EfficientNetB0 base was frozen and only the classification head was optimized using Adam with a learning rate of 1e-3 for five epochs. This allowed the head to adapt to the domain while preserving ImageNet feature representations. In Phase 2, the entire network was unfrozen and fine-tuned end-to-end using Adam with a reduced learning rate of 1e-4 for up to 15 additional epochs. ReduceLROnPlateau (factor=0.3, patience=3) dynamically reduced the learning rate upon validation loss plateaus, and EarlyStopping (patience=5) restored the best weights to prevent overfitting.

4. Experimental Setup

All experiments were conducted on Kaggle Notebook infrastructure equipped with an NVIDIA Tesla P100 GPU (16 GB HBM2 memory) and 13 GB RAM. The software environment comprised Python 3.10, TensorFlow 2.15, NumPy 1.24, scikit-learn 1.3, and seaborn 0.12. Random seeds were fixed at 42 for NumPy and TensorFlow, and at 123 for dataset splitting, ensuring full reproducibility. Table 3 summarizes the hyperparameter configuration.

Table 3. Hyperparameter Configuration

Hyperparameter	Value
Input Image Size	224 × 224 × 3
Batch Size	32
Phase 1 Learning Rate	1e-3
Phase 1 Epochs	5
Phase 2 Learning Rate	1e-4
Phase 2 Max Epochs	15
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy
ReduceLROnPlateau Factor	0.3 (patience=3)
EarlyStopping Patience	5 (restore best weights)
Dropout Rate (head)	0.4, 0.3
Random Seed	42
Train/Val Split	80% / 20%

5. Results

5.1 Training History

Figure 2 presents the training and validation accuracy and loss curves across all 20 epochs (Phase 1: epochs 0–4; Phase 2: epochs 5–19). The dashed vertical line marks the onset of fine-tuning. During Phase 1, validation accuracy improved steadily from approximately 75% to 89%, with training and validation curves closely aligned — indicating no overfitting during the frozen phase. Upon fine-tuning initiation, training accuracy rapidly climbed toward 100% while validation accuracy stabilized at approximately 91–92%. The widening gap between training and validation accuracy in Phase 2 indicates moderate overfitting; however, EarlyStopping with best-weight restoration ensured that the deployed model corresponds to the peak validation checkpoint. Validation loss decreased from 0.61 to approximately 0.33, demonstrating consistent learning progress.

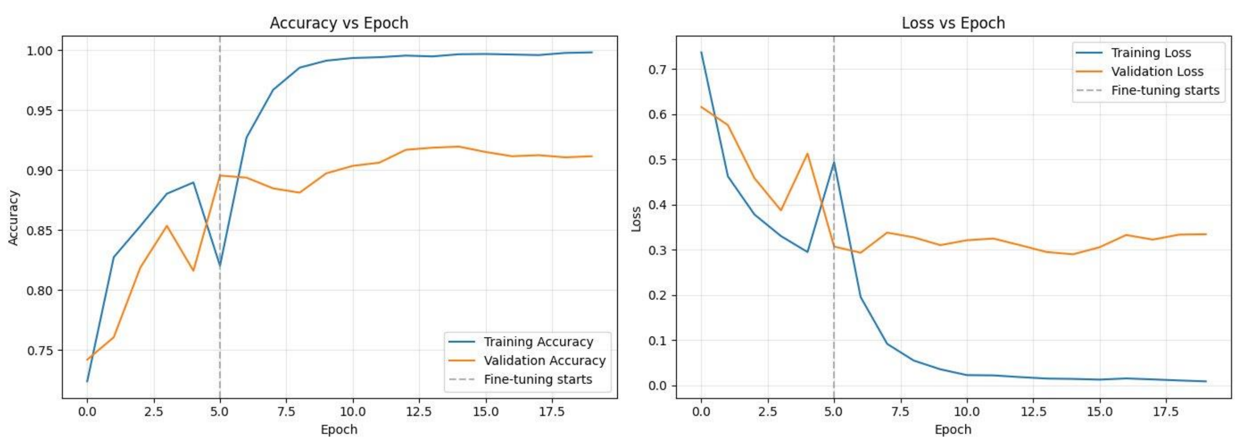

Figure 2. Training and validation accuracy (left) and loss (right) plotted across 20 epochs. The dashed vertical line at epoch 5 indicates the transition from frozen feature extraction (Phase 1) to end-to-end fine-tuning (Phase 2).

5.2 Confusion Matrix

Figure 3 presents the confusion matrices for the test set, displayed as both raw counts and normalized percentages. The model achieved near-perfect classification for the no-tumor (99.25%) and pituitary (99.00%) classes. Glioma yielded a recall of 78.75%, with the majority of errors corresponding to misclassification as no-tumor (9.75%) and meningioma (8.00%). Meningioma demonstrated the lowest recall at 76.00%, with notable confusion toward pituitary (12.25%) and no-tumor (7.75%) categories. These patterns suggest that overlapping MRI intensity profiles and tumor boundary characteristics contribute to glioma-meningioma confusion, consistent with established clinical challenges in differential diagnosis.

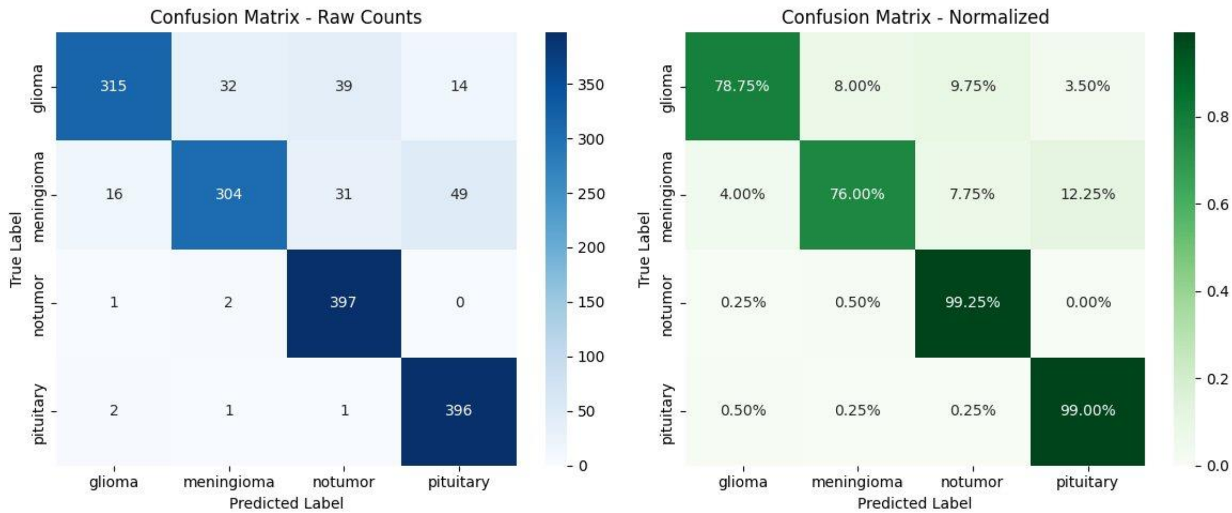

Figure 3. Confusion matrices for the test set. Left: raw prediction counts. Right: row-normalized percentages showing per-class recall. Glioma and meningioma exhibit the highest misclassification rates.

5.3 Comprehensive Performance Metrics

Table 4 reports the complete metric suite computed on the held-out test set. Table 5 provides a direct comparison between the baseline CNN and the proposed EfficientNetB0 model.

Table 4. Per-Class and Aggregate Metrics on Test Set

Class	Precision	Recall	Specificity	F1-Score	AUC
Glioma	0.9357	0.7875	0.9691	0.8553	0.956
Meningioma	0.8916	0.7600	0.9556	0.8207	0.976
No Tumor	0.8455	0.9925	0.9583	0.9131	0.994
Pituitary	0.9512	0.9900	0.9967	0.9702	0.996
Macro Average	**0.9060**	**0.8914**	**0.9699**	**0.8898**	**0.981**
Weighted Average	0.9060	0.9105	—	0.8981	—

Overall Accuracy: 91.05% | *MCC: 0.8822* | *Inference Time: ~2 ms/image* | *Parameters: ~5.3M*

Table 5. Baseline vs. Improved Model Comparison

Metric	Baseline CNN	EfficientNetB0 (Proposed)
Test Accuracy	~73–75%	**91.05%**

Macro F1-Score	~0.70	**0.8981**
Macro Precision	~0.71	**0.9060**
Macro Recall	~0.71	**0.8914**
Macro AUC-ROC	~0.87	**0.981**
MCC	~0.63	**0.8822**
Training Epochs	5	20 (5+15)
Total Parameters	~0.13M	~5.3M

5.4 ROC-AUC Curve

Figure 4 presents the ROC-AUC curves for all four classes under the One-vs-Rest strategy, together with macro-average and micro-average composite curves. The pituitary class attained the highest AUC (0.996), followed by no-tumor (0.994), meningioma (0.976), and glioma (0.956). The macro-average AUC of 0.981 and micro-average AUC of 0.976 both indicate excellent overall discriminative performance across all classes. All curves exhibit substantial separation from the random classifier diagonal, confirming that the model has learned highly informative feature representations for each tumor type.

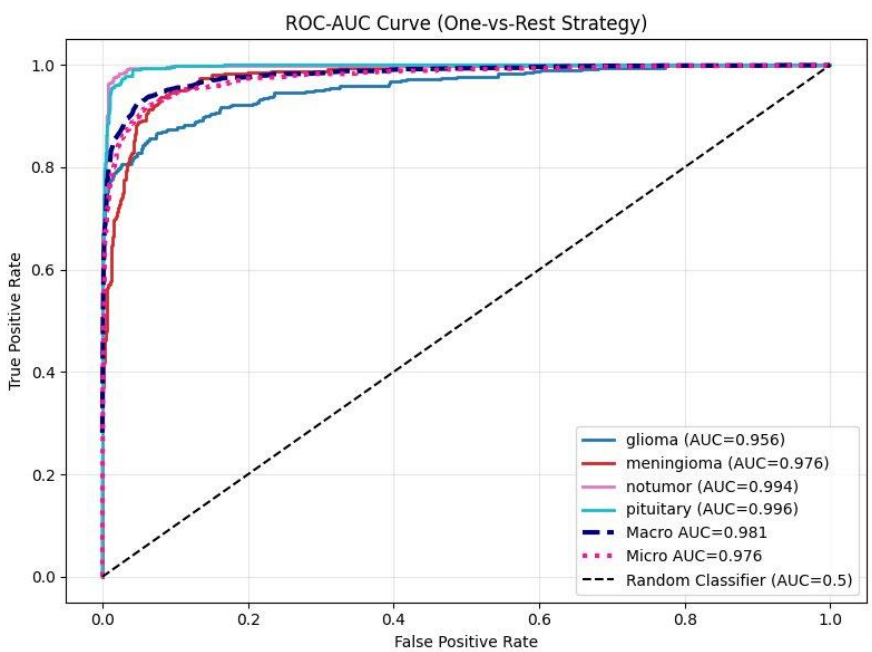

Figure 4. ROC-AUC curves using the One-vs-Rest strategy for each class. The dashed black diagonal represents a random classifier (AUC = 0.5). The macro-average AUC of 0.981 and micro-average AUC of 0.976 confirm strong multi-class discrimination.

5.5 Precision-Recall Curve

Figure 5 depicts the Precision-Recall curves per class. The pituitary class achieved the highest Average Precision (AP = 0.986), followed by no-tumor (AP = 0.984), meningioma (AP = 0.936), and glioma (AP = 0.927). The PR curves are particularly informative for this dataset given the moderate class imbalance between the glioma and no-tumor/pituitary populations. The near-unity precision at low recall levels for all classes confirms that the model can operate in a high-precision regime suitable for clinical pre-screening applications, where false positives are costly.

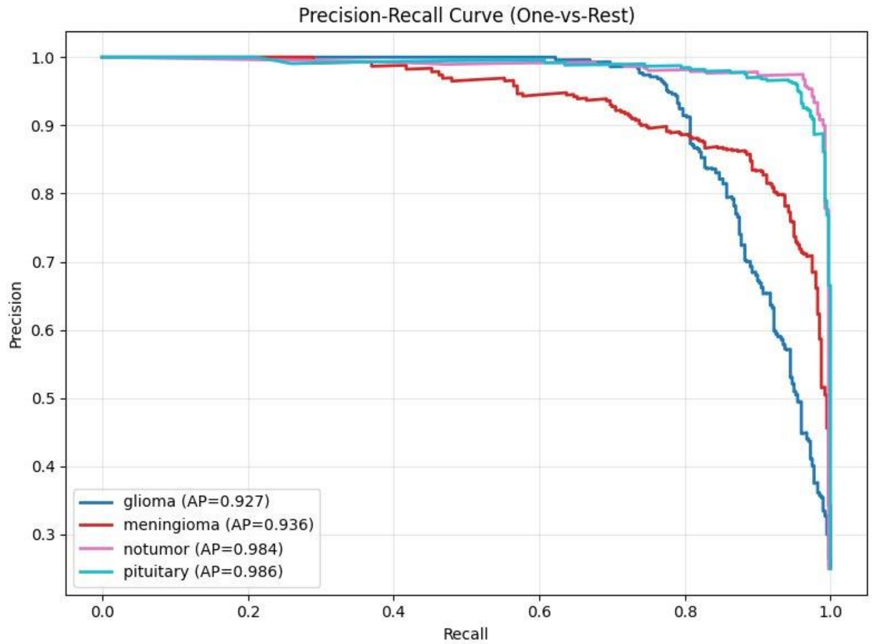

Figure 5. Precision-Recall curves (One-vs-Rest) for each tumor class. Average Precision (AP) scores are reported in the legend. Glioma and meningioma exhibit lower AP scores, reflecting their higher misclassification rates compared to pituitary and no-tumor classes.

5.6 Failure Analysis

Figure 6 presents five representative misclassified examples extracted from the test set. The failure cases reveal several recurring patterns. The first case shows a glioma predicted as pituitary with 65.2% confidence — a notably low confidence score suggesting the model was uncertain, which may flag such cases for human review in a clinical workflow. Cases two through four involve glioma images predicted as no-tumor with high confidence (92–100%), indicating that specific glioma morphologies with diffuse boundaries were visually indistinguishable from normal scans under the learned feature representations. The fifth case depicts a meningioma predicted as pituitary (96.9% confidence), likely due to similar contrast enhancement patterns between the two tumor types in axial plane imaging. These findings suggest that future improvements should focus on glioma boundary delineation and meningioma-pituitary discrimination, potentially through targeted augmentation or multi-scale feature fusion strategies.

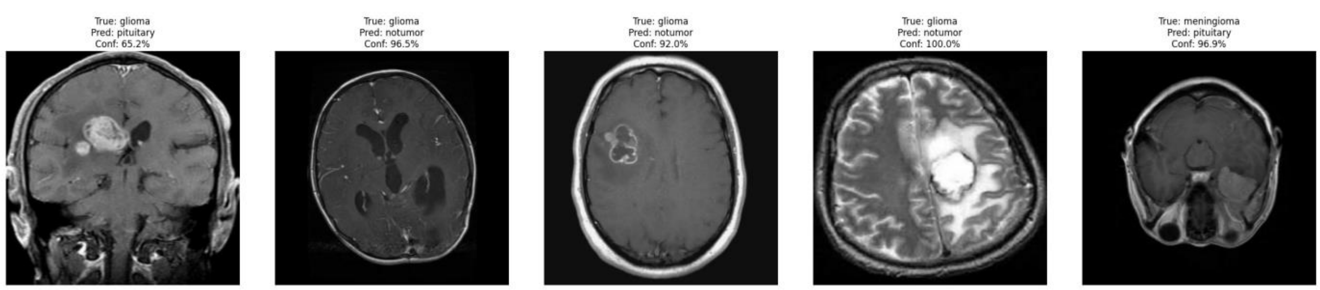

Figure 6. Failure analysis: five representative misclassified test samples. Each panel shows the true label, predicted label, and model confidence. High-confidence errors (92–100%) in glioma-to-notumor misclassifications represent the most clinically significant failure mode.

6. Discussion

The results demonstrate that transfer learning with EfficientNetB0 substantially outperforms the simple CNN baseline across all evaluated metrics. The most significant improvement is observed in the MCC, which increased from approximately 0.63 to 0.8822 — a metric particularly informative for multi-class imbalanced problems, as it accounts for all cells of the confusion matrix simultaneously. The 91.05% test accuracy represents an approximately 16-18 percentage point improvement over the baseline, attributable to the rich feature representations learned from ImageNet pre-training and refined through domain-specific fine-tuning.

The two-phase training strategy proved effective in balancing feature retention and domain adaptation. During Phase 1, the frozen backbone prevented the destruction of general-purpose feature hierarchies, allowing the classification head to converge stably. Phase 2 fine-tuning then enabled task-specific refinement of lower-level features, which is critical for medical image analysis where domain-specific textures — such as tumor heterogeneity and peritumoral edema — differ substantially from natural image statistics.

The disparity in performance between classes is clinically meaningful. The near-perfect recall for pituitary tumors (99.0%) reflects the distinctive anatomical location and compact morphology of pituitary adenomas, which present consistently in the sellar region on sagittal and axial MRI views. The no-tumor class similarly achieved 99.25% recall, as normal brain anatomy exhibits consistent signal characteristics. In contrast, gliomas (78.75% recall) and meningiomas (76.00% recall) present with heterogeneous morphologies, variable enhancement patterns, and overlapping intensity profiles that challenge automated classifiers.

The failure analysis reveals that the majority of high-confidence errors involve glioma images misclassified as no-tumor. This is the most clinically dangerous failure mode, as a missed glioma diagnosis could delay treatment. Future work should investigate the application of attention mechanisms or multi-scale convolutional architectures that explicitly model tumor boundary regions, which are the primary discriminating feature between diffuse gliomas and normal tissue.

The inference time of approximately 2 ms per image on GPU hardware indicates that the model is well-suited for real-time clinical integration. The total parameter count of approximately 5.3 million is modest relative to architectures such as VGG19 (143 million parameters), supporting deployment on edge devices commonly used in resource-constrained clinical environments.

7. Conclusion

This paper presented an EfficientNetB0-based transfer learning approach for four-class brain tumor MRI classification. The proposed model achieved 91.05% test accuracy, macro F1-score of 0.8981, MCC of 0.8822, and macro ROC-AUC of 0.981, substantially outperforming a simple CNN baseline. A systematic two-phase training strategy, comprehensive metric evaluation, and failure analysis were provided to ensure scientific rigor and clinical relevance.

The primary limitation of this work is the moderate misclassification rate for glioma and meningioma cases, which are the two most clinically critical tumor types. Future work will investigate multi-scale feature fusion, attention-guided architectures, and ensemble strategies to address these failure modes. Incorporation of patient clinical metadata and multi-modal imaging data (e.g., T1, T2, FLAIR sequences) may further improve classification robustness. Prospective clinical validation on heterogeneous patient populations from multiple institutions remains a necessary step before deployment in clinical decision support systems.

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